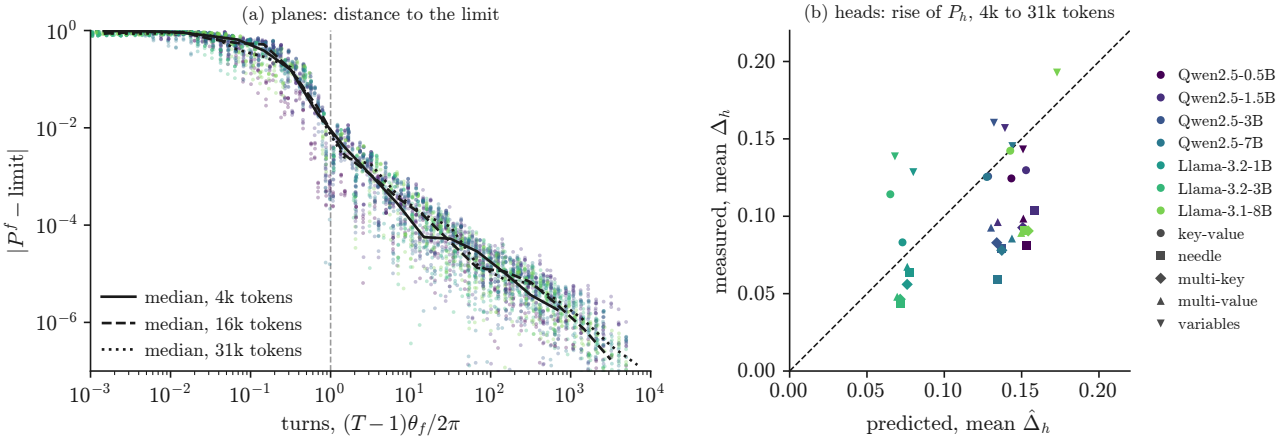

(a) planes: distance to the limit
10^0
10^{-2}
10^{-4}
10^{-6}
$|P^f - \text{limit}|$
median, 4k tokens
median, 16k tokens
median, 31k tokens
10^{-3} 10^{-2} 10^{-1} 10^0 10^1 10^2 10^3 10^4
turns, $(T-1)\theta_f/2\pi$
(b) heads: rise of P_h, 4k to 31k tokens
0.20
0.15
0.10
0.05
0.00
measured, mean Δ_h
0.00 0.05 0.10 0.15 0.20
predicted, mean $\hat{\Delta}_h$
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-1B
Llama-3.2-3B
Llama-3.1-8B
key-value
needle
multi-key
multi-value
variables